H

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE NAME	

BIOLOGY CLASS		REGISTRATION NUMBER	

BIOLOGY

9648/01

Paper 1 Multiple Choice

17 September 2015

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your Biology class, registration number and name above and on the Answer Sheet provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

1 The diagram shows an electron micrograph of a cell.

Which row correctly describes structures P – R?

	P	Q	R
A	degradation of proteins by lysosomal enzymes	substrate level phosphorylation	active replication of genes
B	provision of large surface area for attachment of ribosomes	formation of ATP from light energy	active transcription of genes
C	synthesis of membrane proteins	oxidative decarboxylation	active transcription of genes
D	transport of proteins to Golgi apparatus	modifications of mRNA transcripts	active replication of genes

2 Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it was applied to the roots of the plant. In an attempt to find out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

Which three statements are consistent with the scientist's observations?

1. The toxin is stored in the central vacuole.
2. The toxin cannot cross the membrane of the organelle in which it is stored.
3. The toxin is stored in chloroplast.
4. The toxin is lipid-soluble.
5. The toxin is an enzyme.

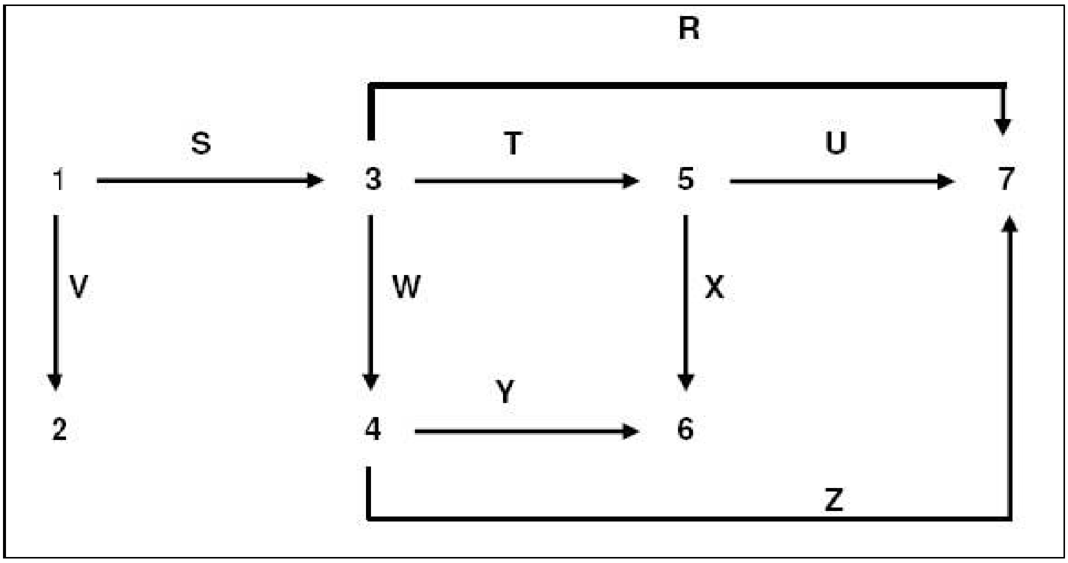
A 1,2 and 5

B 1, 4 and 5

C 2, 3 and 4

D 3, 4 and 5

3 The diagram represents a sequence of reactions taking place in a bacterium in which amino acids are produced from one another by the action of specific enzymes. Numbers 1 to 7 refer to different amino acids. Letters R to Z refer to different enzymes. All the amino acids are essential for survival.



A mutant strain of this bacterium could only survive when provided with amino acids 1, 3 and 6 in its culture medium. The original strain required only amino acid 1. Which enzymes does the mutant bacterium lack?

A S and X only

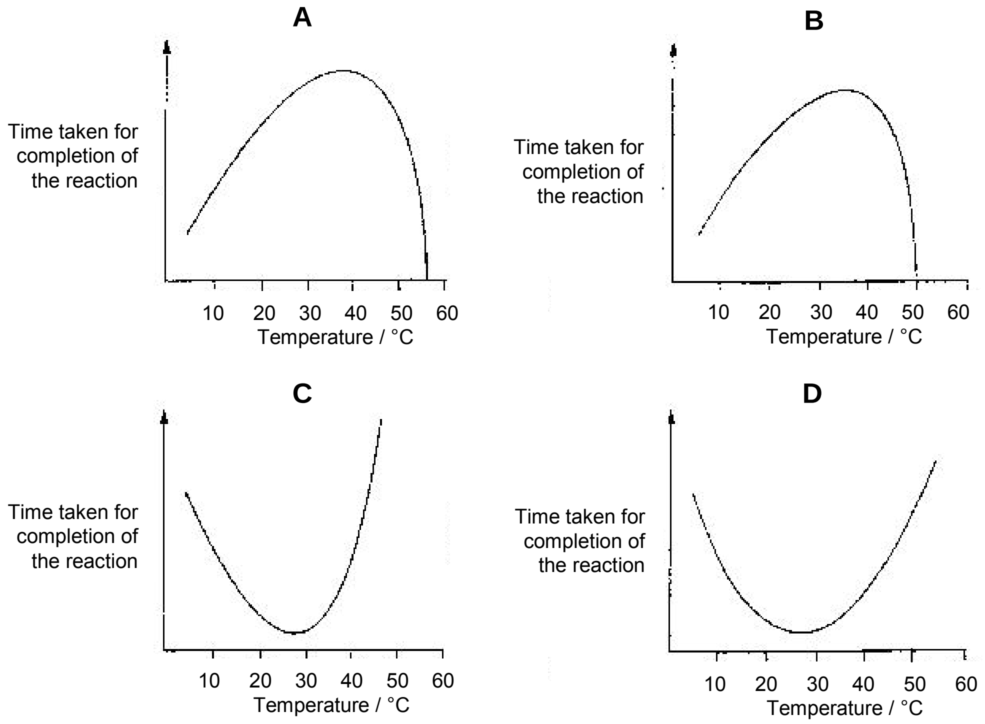
B S, X and Y only

C T, W, X and Y only

D S, T, W, X and Y only

4 In an investigation to determine the effect of temperature on the activity of an enzyme, the time taken for completion of the reaction was recorded.

Which graph shows the results of this investigation?



5 Which of the following statement(s) is/are true?

1. In a neutral solution, most amino acids exists as positively charged compounds.
2. Formation of a peptide bond involves a reaction between an amino group and a carboxyl group.
3. A total of 27 distinct tripeptides can be formed from one valine molecule, one alanine molecule and one leucine molecule.
4. The primary structure of a protein will most likely be preserved when a protein is denatured.

A 1 only

B 2 only

C 2 and 3 only

D 2 and 4 only

6 Which two polysaccharides have the same type(s) of glycosidic linkage types in common?

A amylose and glycogen

B amylose and cellulose

C cellulose and amylopectin

D glycogen and amylopectin

7 Some studies suggest that in patients with Alzheimer's disease, there is a defect in the way the spindle apparatus attaches to the kinetochores. Which stage of mitosis would be affected first?

A anaphase

B metaphase

C prophase

D telophase

8 Which processes would lead to an increase in genetic variation?

1. crossing over between sister chromatids during prophase 1 of meiosis.
2. random distribution of homologous chromosomes to the poles of the cell during anaphase 1 of meiosis
3. random distribution of homologous chromosomes to the poles of the cell during anaphase 2 of meiosis
4. random alignment of homologous chromosomes at the metaphase plate during metaphase 1

A 1 and 4 only

B 2 and 4 only

C 1, 2 and 3 only

D 1, 2 and 4 only

9 In an experiment, bacteria were first grown on a medium containing light nitrogen isotope (^{14}N). They were then transferred to a medium containing heavy nitrogen isotope (^{15}N) and allowed to divide once. The bacteria were subsequently transferred to a medium containing ^{14}N and allowed to divide two more times.

Which row correctly describes the DNA of the bacteria after the experiment?

	Percentage of DNA molecules		
	^{14}N nucleotides only	^{14}N and ^{15}N nucleotides	^{15}N nucleotides only
A	50	25	25
B	75	0	25
C	75	25	0
D	50	50	0

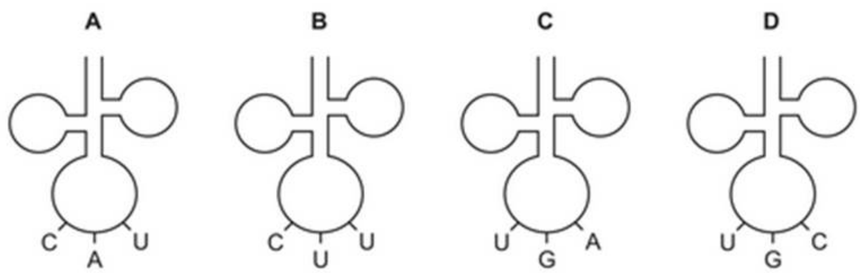
10 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

normal haemoglobin — sickle cell haemoglobin
thr-pro-glu-glu — thr-pro-val-glu

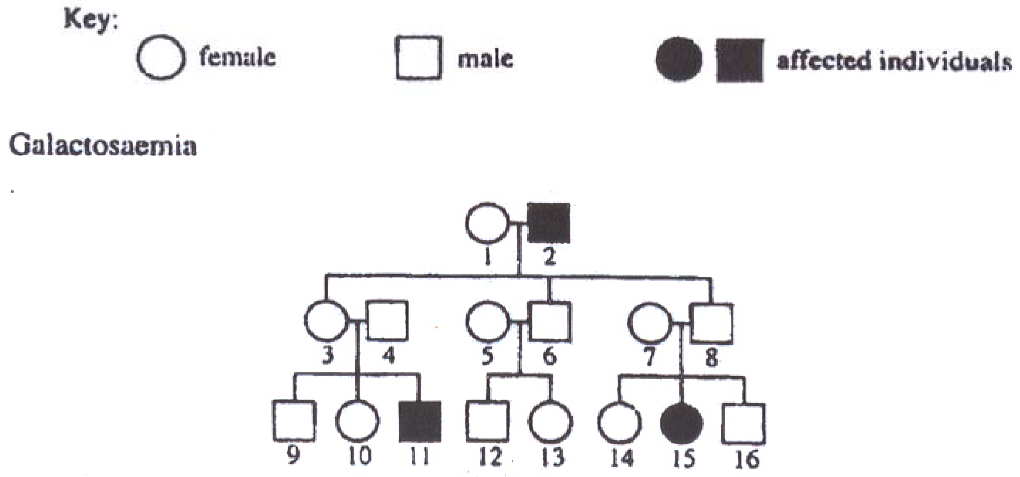
Possible mRNA codons for these amino acids are shown below.

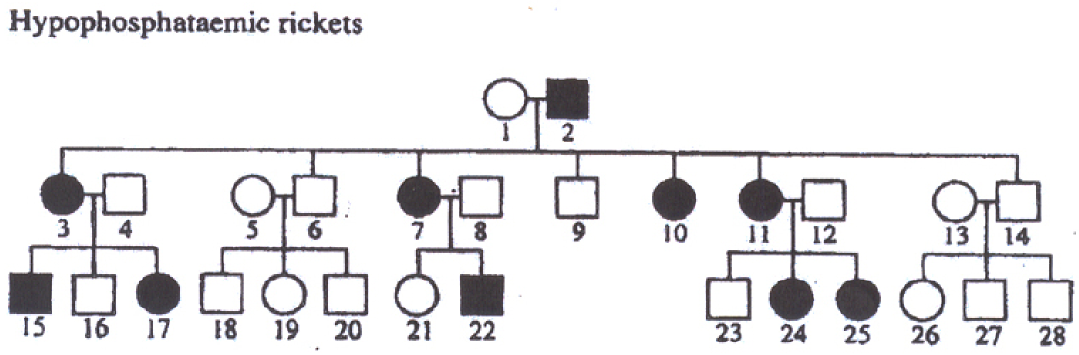
glutamic acid (glu) GAA GAG — proline(pro) CCU CCC
threonine (thr) ACU ACC — valine (val) GUA GUG

Which tRNA molecule is not involved in the formation of this part of the sickle cell haemoglobin?



11 The pedigrees show the inheritance of two genetically transmitted diseases.





Which row correctly describes the The alleles responsible for causing galactosaemia and hypophosphataemic rickets?

A autosomal dominant; sex-linked recessive

B autosomal recessive; sex-linked dominant

C sex-linked dominant; autosomal recessive

D sex-linked recessive; autosomal dominant

12 The feather colours in budgies (a kind of bird) are determined by two pairs of unlinked genes, Y/y and B/b. The genotypes of green, blue and white feathers are Y_B_ and yyB_, and _ _bb (where _ indicates the presence of either the dominant or recessive allele) respectively.

If two heterozygous budgies are mated, what proportion of phenotypes would be expected in the progeny?

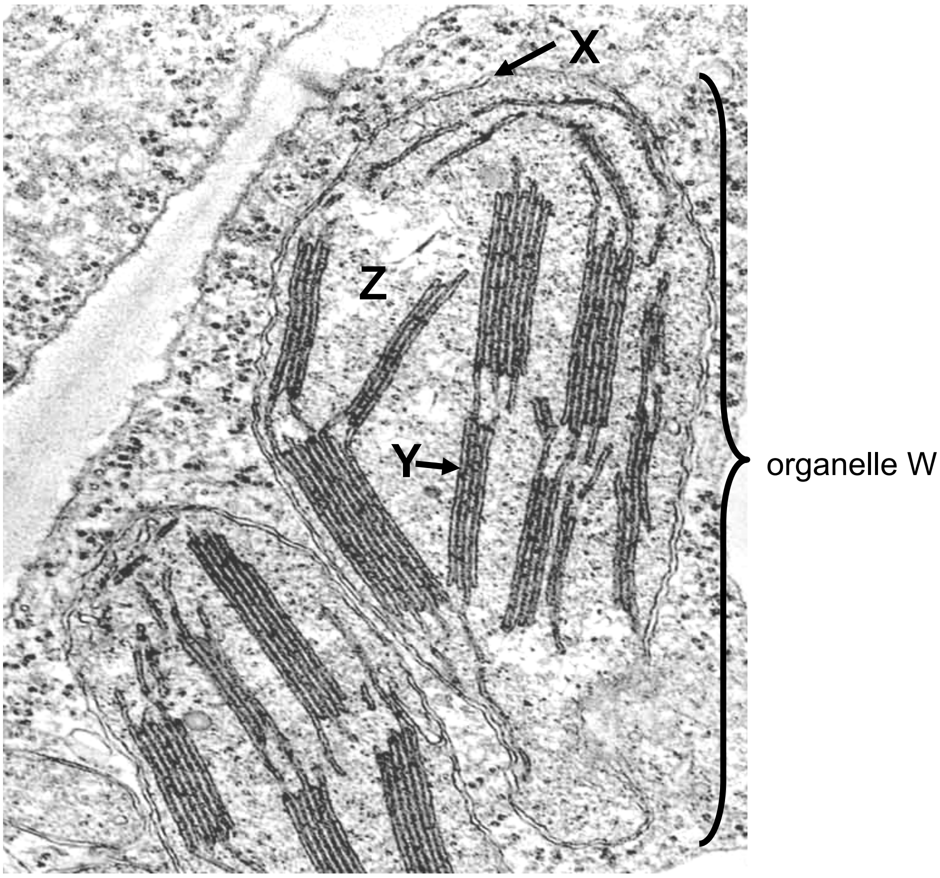
A 9 green feathers : 7 white color feathers

B 9 green feathers : 3 blue feathers : 4 white feathers

C 15 green feathers : 1 blue feathers

D 15 green feathers : 1 white feathers

13 The electron micrograph shows part of a cell containing organelle W.



The table shows the protein composition of various areas in organelle W.

area in organelle W	protein composition (%)
X	6
Y	21
region enclosed by X	6
region Z (between X & Y)	67
TOTAL	100

What does not account for the difference in protein composition between structure Y and region Z?

A presence of ATP synthases in structure Y

B presence of electron carriers in structure Y

C presence of enzymes for Calvin Cycle in region Z

D presence of enzymes for Krebs Cycle in region Z

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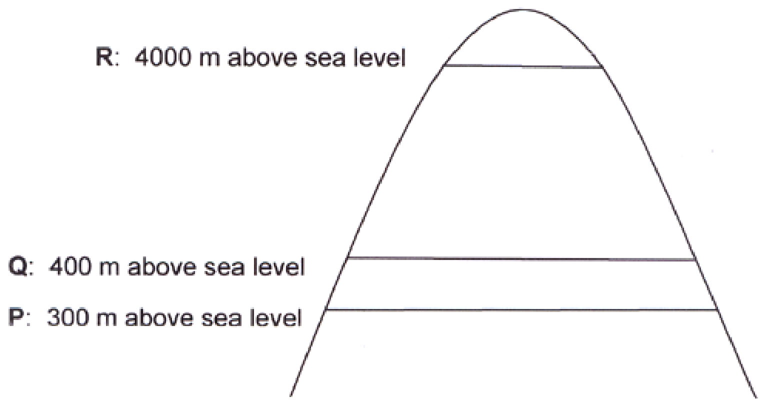
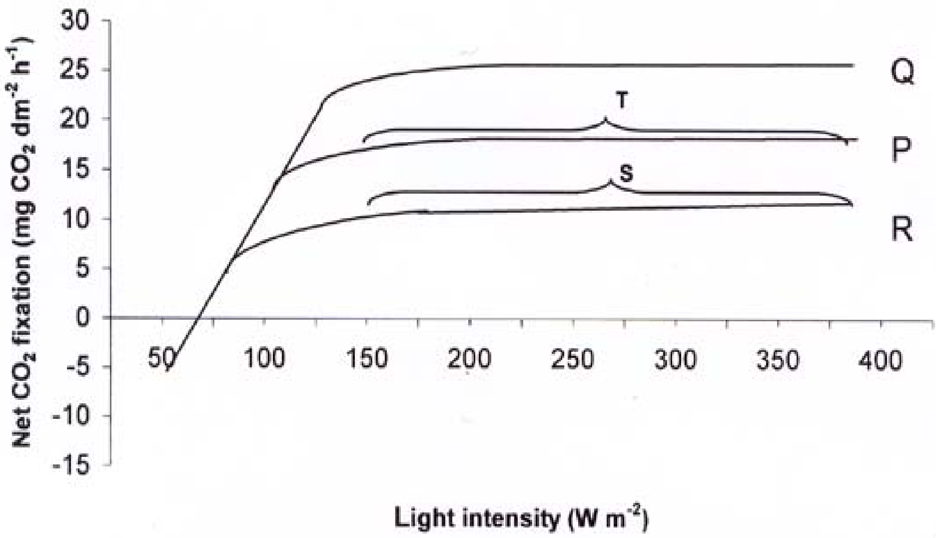
14 The wild orchid species, *Orchii orite*, can be found on Mount Daphne in Western Europe. The diagram shows how Mount Daphne can be divided into several main regions.



The climate at Q in 2004 is the same as that at P before the industrial revolution of the 1850's. The net CO_2 fixation in *Orchii orite* was recorded over a range of light intensities. The following graph shows the experimental data.

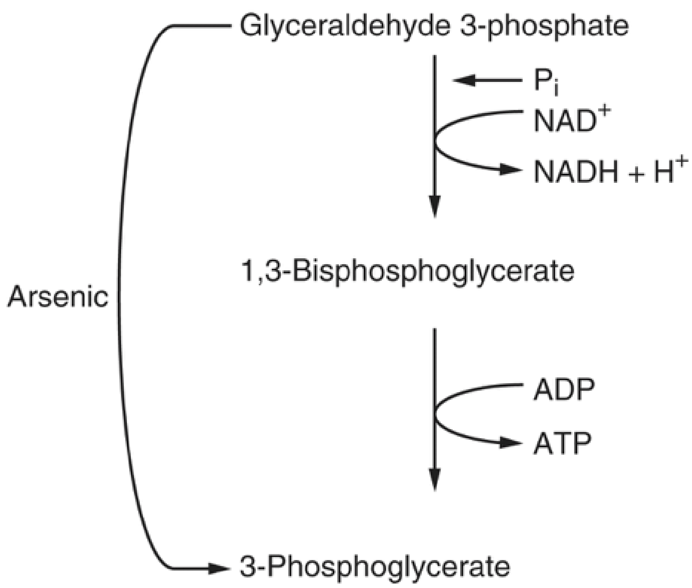


P : 300 m above sea level, pre-industrial revolution (before 1850)
Q : 400 m above sea level, 2004
R : 4000 m above sea level, 2004

What could be the most likely limiting factors at S and T respectively?

	S	T
A	CO_2 concentration	light intensity
B	temperature	CO_2 concentration
C	CO_2 concentration	temperature
D	temperature	water

15 The diagram shows the effect of arsenic on the metabolism of glyceraldehyde-3-phosphate. What is the net yield of ATP molecules from the conversion of two glucose molecules to four pyruvate molecules in the presence of arsenic?



A 0

B 1

C 2

D 4

16 A patient has been exposed to a toxic compound that increases the permeability of mitochondrial membranes to protons. Which metabolic change would be expected in this patient?

A increased ATP levels

B Increased oxygen utilization

C increased ATP synthase activity

D decreased pyruvate dehydrogenase activity

17 Tetrodotoxin, a puffer fish toxin, blocks voltage-gated sodium channels. Black widow spider's venom causes voltage-gated calcium channels to be constantly open. Crotoxin binds irreversibly to acetylcholine receptors.

Which row correctly describes the effect of each toxin on nerve impulse transmission?

	tetrodotoxin	black widow spider's venom	crotoxin
A	blocks transmission of action potential along axon	reduces transmission of impulse across synapse	increases transmission of impulse across synapse
B	increases transmission of impulse across synapse	reduces transmission of impulse across synapse	blocks transmission of action potentials along axon
C	Blocks transmission of action potentials along axon	increases transmission of impulse across synapse	reduces transmission of impulse across synapse
D	reduces transmission of impulse across synapse	blocks transmission action potentials along axon	increases transmission of impulse across synapse

18 When a stimulus arrives at a nerve ending, the plasma membrane becomes depolarised. The events during depolarisation are listed below.

1. K^+ channels open, Na^+ channels close.
2. Na^+ channels open and Na^+ floods in.
3. Potential difference across the plasma membrane is +40 mV.
4. Potential difference across the plasma membrane is -65 mV.

What is the correct order of events?

A 1, 2, 3, 4

B 2, 4, 1, 3

C 3, 4, 1, 2

D 4, 2, 3, 1

19 What would be the best technique for determining the evolutionary relationship among cats, dogs and rats?

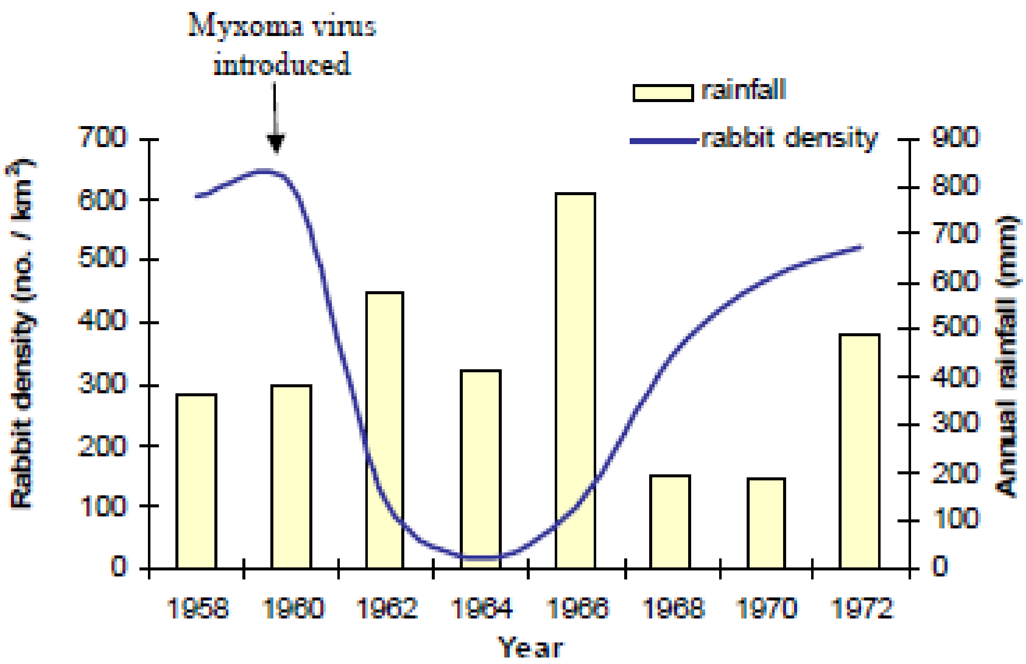
A Compare the anatomical structures of cats, dogs and rats.

B Compare the embryological structures of cats, dogs and rats.

C Compare the genetic information contained in the liver cells of cats, dogs and rats.

D Compare the genetic information contained in the red blood cells of cats, dogs and rats.

20 The graph shows the rabbit density and annual rainfall in a region of France from 1958 to 1972. The rabbit myxoma virus, which causes localized skin tumours, was introduced in this country in 1960.



From the information given, what is the most likely explanation for the change in rabbit density over this period?

A accumulation of spontaneous mutations in the myxoma virus

B differential selection of rabbits by predators

C acquired resistance of some rabbits to the myxoma virus

D none of the above

21 When trypsin converts chymotrypsinogen to chymotrypsin, some molecules of chymotrypsin bind to a repressor, which in turn binds to the operator and prevents further transcription of trypsin gene. This is most similar to which of the following operons?

A *trp* operon during lack of tryptophan

B *trp* operon during abundance of tryptophan

C *lac* operon during lack of lactose

D *lac* operon during abundance of lactose

22 A bacterium, which is resistant to penicillin is transferred to a bacterial colony that lacks the fertility factor. The rest of the colony does not become resistant to penicillin. However, the penicillin-resistant bacterium started to exhibit other phenotypic characteristics, including secretion of novel protein. Which process is not likely to account for this change?

A conjugation

B mutation

C transduction

D transformation

23 Antibiotic resistance is a growing problem that makes treating common bacterial infections more difficult. Which mechanism(s) could account for a bacterium's ability to increase its genetic variation and thus adapt itself to different antibiotics?

1. binary fission
2. conjugation
3. transduction

A 2 only

B 1and 3 only

C 2 and 3 only

D 1, 2 and 3

24 What statements are true about G-protein signaling?

1. During activation of G-protein, subunit of the G-protein dissociates from the activated G-protein to activate adenylyl cyclase
2. During inactivation of G-protein, the active α subunit is inactovated by the hydrolysis of the bound GTP caused by GTPase
3. Testosterone can bind to the cell membrane receptor to activate G-protein
4. The ratio of G-protein coupled receptor to G-protein is 1:1.

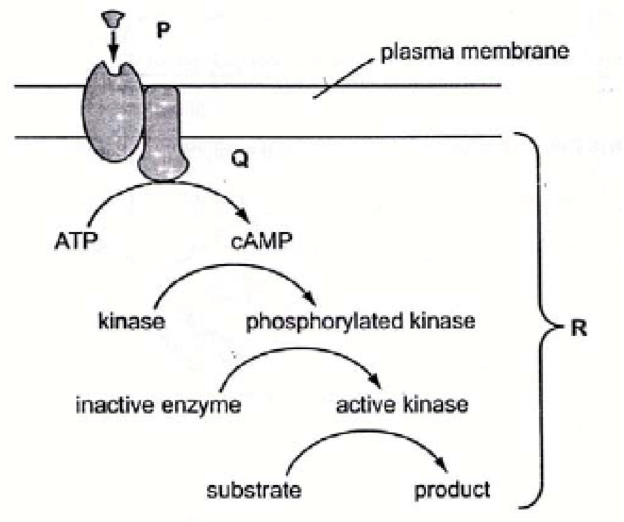
A i only

B ii only

C i, ii, and iii only

D i, ii, iii, and iv.

25 The diagram shows an example of cell signaling.



Which row correctly describes P,Q and R?

	P	Q	R
A	amplification	ligand-receptor interaction	phosphorylation
B	ligand-receptor interaction	amplification	phosphorylation
C	ligand-receptor interaction	transduction	amplification
D	transduction	phosphorylation	amplification

26 The haemaglutinin glycoprotein only allows influenza to infect cells with sialic acid receptors. To increase the host range of influenza in a laboratory setting, scientists often modify these viruses by artificially enclosing the viral nucleocapsid by a viral envelope with VSV-G glycoproteins that are obtained from the vesicular stomatitis virus. VSV-G glycoprotein binds to LDL receptor, which is found ubiquitously on the cell membranes of many cell types.

What can the scientists expect the resulting progeny viruses to have?

	viral envelope	viral genome
A	VSV-G glycoprotein	VSV-G gene
B	VSV-G glycoprotein	haemaglutinin gene
C	haemaglutinin glycoprotein	VSV-G gene
D	Haemaglutinin glycoprotein	haemaglutinin gene

27 How can viruses cause disease in animals?

1. They inhibit normal host cell DNA, RNA, or protein synthesis.
2. They deplete the host cell of cellular materials essential for metabolic functions.
3. Their viral proteins and glycoproteins on the surface membrane of host cells cause them to trigger host's immune responses.
4. They disrupt and inactivate the oncogenes of the host cell causing uncontrolled cell division.

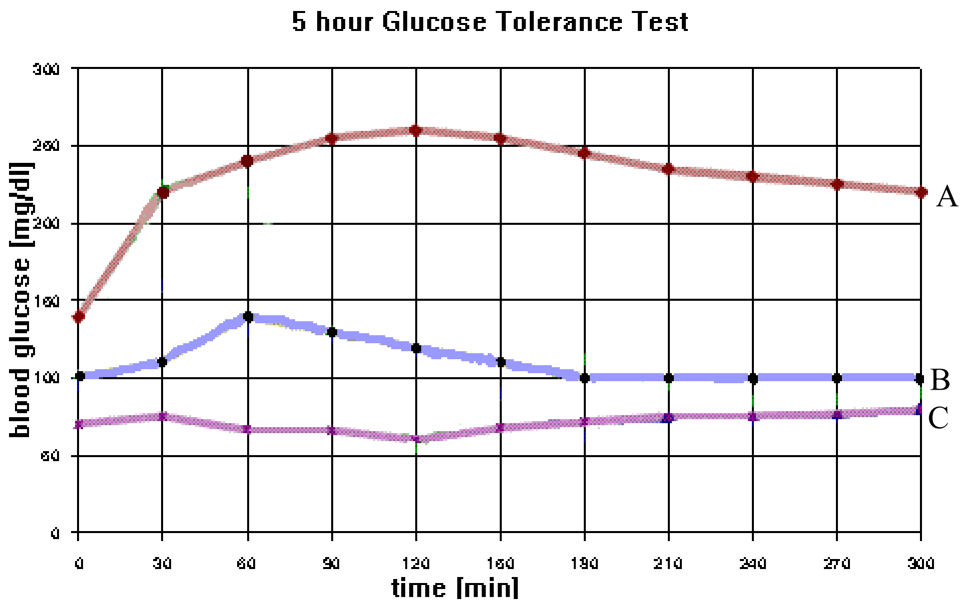
A 1 and 2 only

B 1, 2 and 3 only

C 1, 2 and 4 only

D 1, 2, 3 and 4

28 The five hour glucose tolerance test involves drinking a 100g glucose solution, and then measuring the blood glucose level over five hours. Persons A, B and C took the test. The graph shows the results of their glucose tolerance test.



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What cannot be concluded from the graph?

A In a normal healthy individual, it takes three hours for insulin to restore normal blood glucose levels.

B Lowering of blood glucose level in person A is most probably brought about by utilisation of glucose in respiration.

C The beta cells of the islets of Langerhans in person A are non-functional.

D The pancreas released insulin 60 minutes after it had detected the rise in blood glucose level.

29 Which statements are true about signal transduction pathways?

1. They enable different cells to respond appropriately to the same signal.
2. They help cells use up phosphate generated by ATP breakdown.
3. They can amplify a signal.
4. A single ligand can lead to different signal transduction pathways, resulting in different cellular responses.

A 1, 2 and 3

B 1, 3 and 4

C 2 and 4

D 1 and 3

30 What is not a factor that may increase the chance of cancerous growth?

A daily exposure to acrylamide and benzene at work

B infection by influenza virus

C inheritance of p53 gene mutation

D smoking after every meal

31 The human epidermal growth factor receptor 2 is encoded by the HER2 gene, a proto-oncogene.

Which genetic change would convert the proto-oncogene into an oncogene?

A amplification of the promoter that controls the proto-oncogene

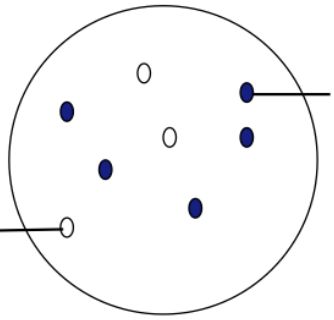
B amplification of the proto-oncogene

C point mutation in an intron within the proto-oncogene

D translocation of the proto-oncogene to a region near the centromere

32 pACT-A is a 7.25kb plasmid cloning vector that is often used in creating recombinant DNA. It contains two selectable marker genes, *TetR* and *lac*Z. The multiple cloning site (MCS) of the cloning vector is located within the *lac*Z gene.

Using *BamHI* as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS of plasmid pACT-A and the recombinant plasmid is intorduced into *E. coli* cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



Colonies X and Y are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *BamHI*. Which row correctly shows the fragments obtained?

	Colony X	Colony Y
A	9.56kb fragment	7.25kb fragment
B	7.25kb and 2.31 kb fragments	7.25kb fragment
C	7.25kb and 0.34kb fragments	7.25kb and 2.31kb fragments
D	7.25kb and 0.34kb fragments	7.25kb and 0.34kb fragments

33 Which pair of primers should be used in polymerase chain reaction to amplify the DNA flanked by the two indicated sequences?

5' – GGAATTCGT -- // -- AATGCTACC – 3'

A AATGCTACC & GGAATTCGT

B ACGAATTCC & AATCGTACC

C ACGAATTCC & GGTAGCAAT

D GGTAGCATT & GGAATTCGT

34 What are normal functions of stem cells in a living human?

	normal functions of human stem cells		
A	differentiating into many kinds of cells in a 3-5 day old human embryo	producing insulin in pancreas damaged by type 1 diabetes	cells that can differentiate into bone cells in the skeleton
B	differentiating into many kinds of cells in a 3-5 day old human embryo	producing red blood cells worn out by normal wear and tear	cells that can differentiate into cardiac muscle cells in the heart
C	differentiating into one kind of a cell in a 3-5 day old human embryo	producing dopamine in the brains of people with Parkinson's	cells that can differentiate into cartilage cells in the joints
D	differentiating into one kind of a cell in a 3-5 day old human embryo	producing differentiated cells that can be used to screen new drugs	cells that can differentiate into nerve cells in the brain

35 Which pair of statements are ethical concerns arising from the Human Genome Project?

1. scientists tracing migration of different population groups based on maternal inheritance
2. genetic counsellors giving advice to people who are genetically pre-disposed to risks
3. parents choosing to abort foetuses with minor disorders based on genetic testing results
4. scientists developing tests for only some disease-causing genes
5. insurance company charging lower rates to people with genetic disposition to fewer diseases

A 1 and 4

B 2 and 5

C 3 and 4

D 3 and 5

36 One way to treat β-thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal, dominant allele for β-globin to the patient's cells.

What would ensure that the normal gene is passed on to the next generation?

A using a retrovirus to introduce the normal β-globin gene into bone marrow cells

B using a retrovirus to introduce the normal β-globin gene into an egg cell

C using an adenoviral vector to introduce the normal β-globin gene into bone marrow cells

D using an adenoviral vector to introduce the normal β-globin gene into an egg cell

37 In plant tissue culture, there are five main stages involved, starting from getting the explant to the acclimatisation of the plant.

Which of the following plant growth regulator ratios are used in the establishment of plantlets?

	relative amount of cytokinins	relative amount of auxins	ratio of plant growth regulator concentration
1	low	low	optimum ratio dependant on type of plant
2	low	intermediate	intermediate auxin: cytokinin
3	high	low	high cytokinin:auxin
4	low	high	high auxin:cytokinin

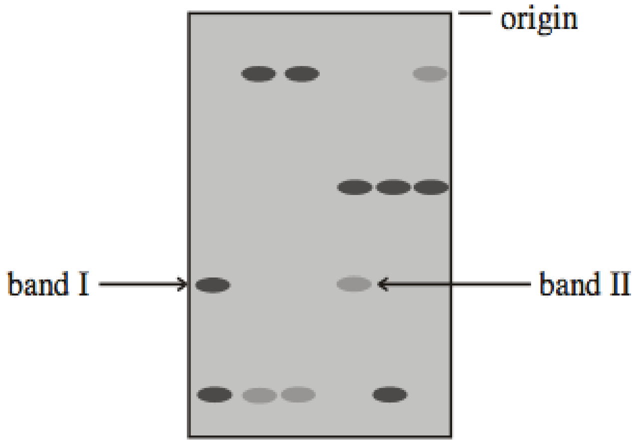
A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

38 The diagram shows the results of DNA profiling using gel electrophoresis.



What conclusion can be drawn about the DNA in bands I and II?

A The DNA in the two bands had the same base sequence.
B The DNA in the two bands had the same ratio of bases.
C The DNA in the two bands came from the same source.
D None of the above

39 Erythropoietin (EPO) is a small signaling molecule produced by certain kidney cells. It stimulates the production and differentiation of red blood cells. A scientist is interested to clone the cDNA of EPO receptor (EPO-R) from mouse and express it in suitable human host cells. The first step in library construction is cDNA synthesis.

Which combination is required for synthesizing a collection of cDNAs containing copies of EPO-R cDNA?

	DNA ligase	DNA polymerase	RNA polymerase	reverse transcriptase
A	√	×	√	×
B	√	√	×	×
C	×	×	√	√
D	×	√	×	√

40 What is an example of genetically modified organisms?

A Durians that ripen slower, due to anti-sense technology.

B Cows that grow to adult sizes quickly due to Bovine Somatotrophin injection

C High yielding rice that are flood resistant, due to intensive self-pollination over generations.

D Cows that grow to abnormal size due to inbreeding.